EDC · MOCK

EDC mock A

Practice questions · Day 27 · Thu 19 Nov · pack EX1

How to use this: Sit this paper two days before the EDC exam, timed (2 hours), with your hard-copy notes and a calculator, and do not look at the marking guide. Then mark it strictly with the guide below. The paper is modelled on the practice test (open book, formula table first, MCQ concepts and then workshop-style calculations). It has been scaled up to a full 2-hour final covering the whole course. About 43% of the marks are on diodes and BJTs and about 57% on FETs, multistage amplifiers, op-amps and power amplifiers. Every circuit is described in words, so you do not need a figure. **Draw each circuit yourself before you start the question.** That takes about a minute and it earns the “redraw” marks.

Instructions to candidates

- This is an open-book examination. No internet access.
- Duration: **2 hours**. There are **10 questions**. Answer all of them. Total: **82 marks**.
- The numbers in square brackets are the marks for each part.
- Show detailed calculations: write the formula in symbols, then substitute, then give the answer with units.
- Unless a question says otherwise: Si BJTs have $V_{BE,on} = 0.7$ V and $V_{CE,sat} = 0.2$ V, $V_T = 26$ mV at room temperature, and all coupling and bypass capacitors are short circuits at signal frequency and open circuits at DC.
- A sensible pace is about **1.4 minutes per mark**. Keep 5 minutes at the end for checking.

> **FORMULA TABLE**

Topic	Formulas
Diodes	$I_D = I_S\left(e^{V_D/nV_T}-1\right) \approx I_S e^{V_D/nV_T}$; $V_T = kT/q$; $\Delta V_D = nV_T \ln(I_2/I_1) = 2.3\,nV_T \log_{10}(I_2/I_1)$; $r_d = nV_T/I_D$; Si knee 0.7 V, Ge 0.3 V
Rectifier and Zener	$V_r = V_P/(fCR)$; $V.R. = \dfrac{V_{NL}-V_{FL}}{V_{FL}} \times 100\%$; Zener shunt regulator: $I_S = I_Z + I_L$, $I_S = (V_i - V_Z)/R_S$, $P_Z = V_Z I_Z$
BJT DC	$I_C = \beta I_B$, $I_E = (\beta+1)I_B$, $\alpha = \beta/(\beta+1)$; voltage divider: $V_{Th} = \dfrac{R_2}{R_1+R_2}V_{CC}$, $R_{Th} = R_1 \parallel R_2$, $I_B = \dfrac{V_{Th}-V_{BE}}{R_{Th}+(\beta+1)R_E}$; approximate method valid if $\beta R_E \geq 10R_2$
BJT AC (r_e model)	$r_e = 26\ \text{mV}/I_E$; $g_m = I_C/V_T$; $r_\pi = \beta r_e$; CE (bypassed): $Z_i = R_1 \parallel R_2 \parallel \beta r_e$, $Z_o = R_C \parallel r_o$, $A_v = -(R_C \parallel r_o)/r_e$; emitter follower: $Z_b = \beta r_e + (\beta+1)R_E$, $Z_o = R_E \parallel (r_e + R_s'/\beta)$, $A_v = R_E/(R_E+r_e)$
Loading	$A_{v_L} = A_{v_{NL}}\dfrac{R_L}{R_L+Z_o}$; $A_{v_s} = \dfrac{Z_i}{Z_i+R_s}A_{v_L}$; cascade: $A_{v_T} = A_{v1}A_{v2}\cdots$ with each stage loaded by the next stage's Z_i
JFET and D-MOSFET	$I_D = I_{DSS}\left(1-\dfrac{V_{GS}}{V_P}\right)^2$; $V_{GS} = V_P\left(1-\sqrt{I_D/I_{DSS}}\right)$; $g_{m0} = \dfrac{2I_{DSS}}{\lvert V_P\rvert}$; $g_m = g_{m0}\left(1-\dfrac{V_{GS}}{V_P}\right)$; $r_d = 1/y_{os}$
E-MOSFET	$I_D = k\big(V_{GS}-V_{GS(Th)}\big)^2$; $k = \dfrac{I_{D(on)}}{\big(V_{GS(on)}-V_{GS(Th)}\big)^2}$; $g_m = 2k\left(V_{GSQ}-V_{GS(Th)}\right)$; saturation needs $V_{DS} \geq V_{GS}-V_{GS(Th)}$

Topic	Formulas
FET amplifiers	CS (R_S bypassed): $Z_i = R_G$ (or $R_1 \parallel R_2$), $Z_o = r_d \parallel R_D$, $A_v = -g_m(r_d \parallel R_D)$; source follower: $A_v = \dfrac{g_m(r_d \parallel R_S)}{1 + g_m(r_d \parallel R_S)}$
Differential and op-amps	$V_d = V_{i1} - V_{i2}$, $V_c = \frac{1}{2}(V_{i1} + V_{i2})$; $V_o = A_d V_d + A_c V_c$; $\mathrm{CMRR} = A_d/A_c$, $\mathrm{CMRR(dB)} = 20\log_{10}(A_d/A_c)$; BJT pair (single-ended): $A_d = \dfrac{R_C}{2r_e}$, $A_c = \dfrac{\beta R_C}{r_i + 2(\beta+1)R_E}$; inverting: $V_o = -\dfrac{R_f}{R_1}V_1$; non-inverting: $V_o = \left(1 + \dfrac{R_f}{R_1}\right)V_1$; summing: $V_o = -\sum \dfrac{R_f}{R_k}V_k$
Power amplifiers	Class A series-fed: $P_i = V_{CC}I_{CQ}$, $P_o = I_{c(rms)}^2 R_C = \dfrac{I_{c(p)}^2 R_C}{2}$, $\eta = P_o/P_i$; class B push-pull: $P_i = V_{CC}\dfrac{2}{\pi}I_{(p)}$, $P_o = \dfrac{V_{L(p)}^2}{2R_L}$, $P_{o(max)} = \dfrac{V_{CC}^2}{2R_L}$, $P_{i(max)} = \dfrac{2V_{CC}^2}{\pi R_L}$, $P_{2Q(max)} = \dfrac{2V_{CC}^2}{\pi^2 R_L}$; transformer: $R'_L = a^2 R_L$

Question 1 [10 marks]

Circle the correct answer in each multiple-choice question (I to X). [1 mark each]

I. When the reverse bias across a pn junction is increased, the depletion region:

- a. widens, and the junction capacitance decreases
- b. narrows, and the junction capacitance increases
- c. widens, and the junction capacitance increases
- d. does not change, because no current flows in reverse bias

II. A bridge rectifier with a capacitor filter feeds a fixed load. If the filter capacitance is doubled, the peak-to-peak ripple voltage:

- a. doubles
- b. roughly halves
- c. falls to zero
- d. does not change, because ripple depends only on frequency

III. In a Zener shunt regulator with a fixed series resistor R_S, the Zener is most likely to drop out of regulation when:

- a. the supply voltage is at its maximum and the load current is zero
- b. the supply voltage is at its maximum and the load current is at its maximum
- c. the supply voltage is at its minimum and the load current is at its maximum
- d. the load is disconnected

IV. An NPN BJT is measured with $V_{BE} = 0.7$ V and $V_{BC} = +0.5$ V. The transistor is in:

- a. the active region
- b. cut-off
- c. saturation
- d. the reverse-active region

V. A bypass capacitor is connected across R_E in a voltage-divider CE amplifier. Compared with the unbypassed amplifier:

- a. the Q-point moves and the gain falls
- b. the Q-point is unchanged and $|A_v|$ rises from about R_C/R_E to about R_C/r_e
- c. the Q-point is unchanged and the input impedance rises
- d. the Q-point moves to saturation

VI. An n-channel JFET has $V_{GS} = V_P$. Its drain current is:

- a. approximately zero for any V_{DS}
- b. equal to I_{DSS}
- c. equal to $I_{DSS}/4$
- d. set only by the external drain resistor

VII. Which device can be operated with either a positive or a negative gate-to-source voltage and still conduct?

- a. n-channel JFET
- b. n-channel enhancement-type MOSFET
- c. n-channel depletion-type MOSFET
- d. p-channel enhancement-type MOSFET

VIII. For a JFET, the transconductance g_m is largest when:

- a. $V_{GS} = V_P$
- b. $V_{GS} = 0$ V
- c. $V_{GS} = V_P/2$
- d. $I_D = I_{DSS}/4$

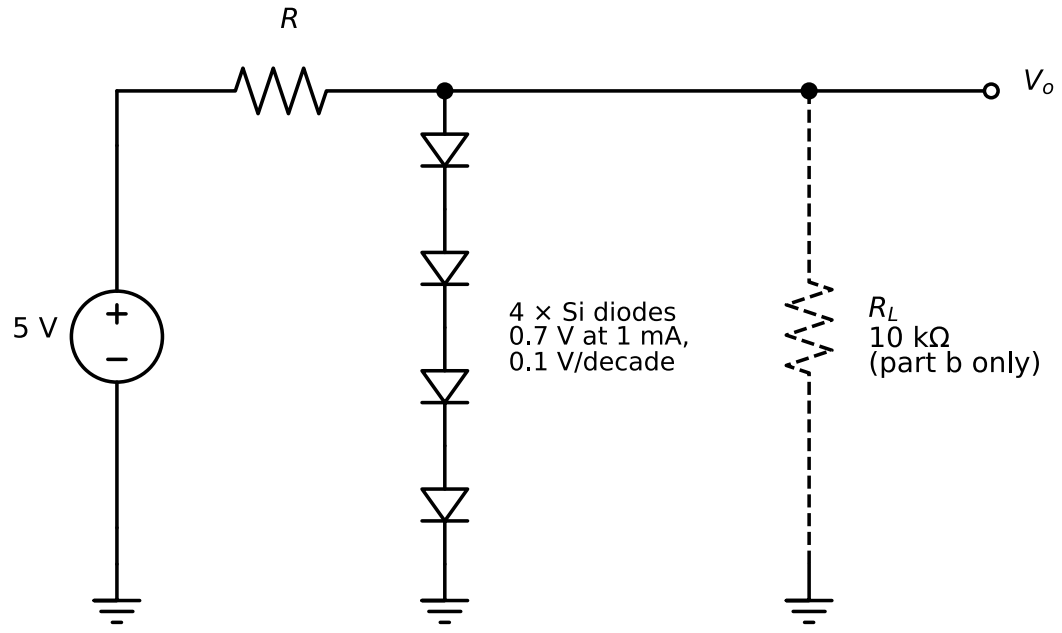
IX. In an ideal op-amp inverting amplifier with negative feedback, the inverting input terminal:

- a. is at virtually the same voltage as the non-inverting input, which is grounded, and it draws no current
- b. is at the output voltage
- c. carries the full input current into the op-amp
- d. is at the input voltage V_1

X. Crossover distortion in a class B push-pull output stage is caused by:

- a. the output transformer saturating
- b. the transistors' β being unequal
- c. the efficiency exceeding 78.5%
- d. both transistors being off while $|v_{in}|$ is below about 0.7 V

Question 2 [8 marks]



Question 2 circuit: 5 V supply, R, four series diodes, optional 10 kOhm load

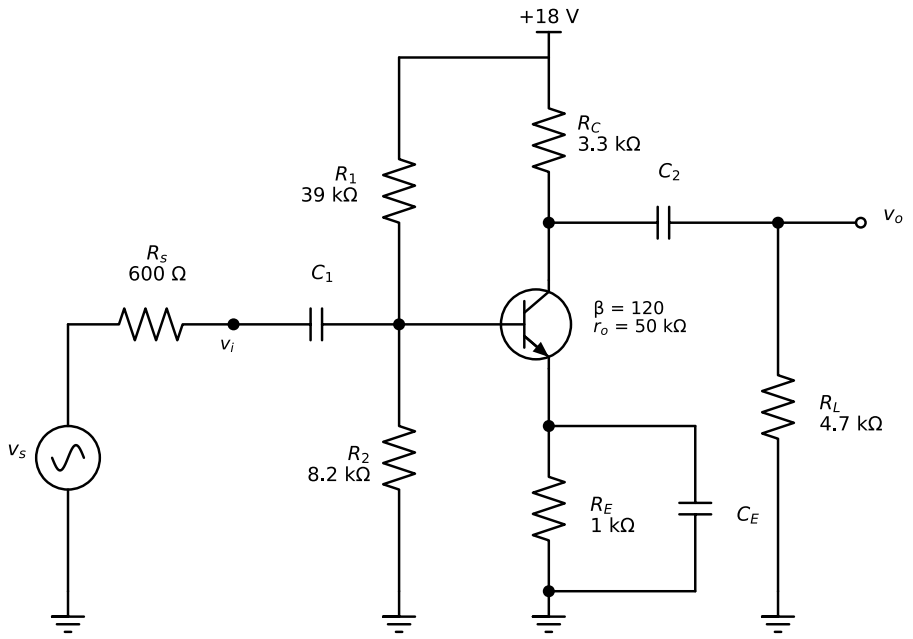
A 5 V DC supply feeds a resistor R. The other end of R is the output node. Between the output node and ground there is a string of **four identical silicon diodes in series**, all forward biased (anode towards the output node). The output V_o is the voltage across the whole string. Each diode has a forward drop of 0.7 V at 1 mA, and this drop changes by 0.1 V for every decade change in current.

a. Design the circuit for $V_o = 2.6$ V with no load: find the diode current and the value of R. [3 marks]

b. A load resistor $R_L = 10\ \text{k}\Omega$ is now connected from the output node to ground, in parallel with the diode string. Keep R at your value from (a). Find the new V_o and the new diode current by iteration. [3 marks]

c. With no load connected, find the incremental resistance of one diode at the design point. Use it to estimate the change in V_o when the supply changes by ± 0.5 V. [2 marks]

Question 3 [13 marks]



Question 3 circuit: voltage-divider CE amplifier with bypassed R_E, source R_s and load R_L

A voltage-divider-biased common-emitter amplifier uses an NPN BJT with $\beta = 120$ and $r_o = 50\ \text{k}\Omega$.

- $V_{CC} = 18$ V. $R_1 = 39\ \text{k}\Omega$ from V_{CC} to the base, and $R_2 = 8.2\ \text{k}\Omega$ from the base to ground.
- $R_C = 3.3\ \text{k}\Omega$ from V_{CC} to the collector, and $R_E = 1\ \text{k}\Omega$ from the emitter to ground. R_E is fully bypassed by a capacitor C_E.
- The signal source v_s has internal resistance $R_s = 600\ \Omega$ and is capacitively coupled to the base.
- The output is taken from the collector through a coupling capacitor to a load $R_L = 4.7\ \text{k}\Omega$.

a. State whether the approximate method may be used. Then use the **exact (Thevenin) method** to find V_{Th}, R_{Th}, I_B, I_C, I_E and V_{CE}. Confirm that the BJT is in the active region. [5 marks]

b. Find r_e. Then find Z_i, Z_o and the no-load voltage gain $A_v = v_o/v_i$, including r_o. [4 marks]

c. Find the loaded gain $A_{v_L} = v_o/v_i$ with R_L connected, and the overall gain $A_{v_s} = v_o/v_s$. Find the peak output voltage for $v_s = 10$ mV peak. State which of R_s and R_L costs more gain in this circuit, and why. [4 marks]

Question 4 [9 marks]



An emitter follower uses an NPN BJT with $\beta = 100$ (r_o may be ignored).

- $V_{CC} = 12$ V is connected directly to the collector.
- $R_B = 220\ \text{k}\Omega$ is connected from V_{CC} to the base.
- $R_E = 2.2\ \text{k}\Omega$ is connected from the emitter to ground.
- The source v_s, with $R_s = 1\ \text{k}\Omega$, is capacitively coupled to the base.
- The output is taken from the emitter through a coupling capacitor to a load $R_L = 1\ \text{k}\Omega$.

a. Find I_B, I_E, V_E and V_{CE}. [2 marks]

b. Find r_e. Find Z_i with no load, and again with R_L connected. [2 marks]

c. Find the output impedance Z_o seen by R_L, including the effect of R_s. [2 marks]

d. Find $A_{v_L} = v_o/v_i$ and $A_{v_s} = v_o/v_s$. [2 marks]

e. In one sentence, explain why this stage is useful even though its gain is less than 1. [1 mark]

Question 5 [8 marks]



An n-channel JFET with $I_{DSS} = 10$ mA, $V_P = -4$ V and $y_{os} = 20\ \mu\text{S}$ is used in a **self-bias** common-source amplifier.

- $V_{DD} = 20$ V. $R_D = 2.2\ \text{k}\Omega$ from V_{DD} to the drain.
- $R_S = 680\ \Omega$ from the source to ground, bypassed by a capacitor C_S.
- $R_G = 1\ \text{M}\Omega$ from the gate to ground. The input is capacitively coupled to the gate. The output is taken from the drain through a capacitor (no load).

a. Write the equation for I_D by combining Shockley's equation with the bias line. Solve the quadratic, show **both** roots, and choose the correct one with a reason. State I_{DQ} and V_{GSQ}. [3 marks]

b. Find V_S, V_D and V_{DS}. Check that the JFET is in the saturation (pinch-off) region. [2 marks]

c. Find g_m, r_d, Z_i, Z_o and A_v. [2 marks]

d. C_S is removed. Taking r_d as very large, derive and evaluate the new A_v. [1 mark]

Question 6 [7 marks]

Question 6 circuit: E-MOSFET voltage-divider bias

An n-channel enhancement-type MOSFET has $V_{GS(Th)} = 3$ V and $I_{D(on)} = 5$ mA at $V_{GS(on)} = 8$ V. It is biased with a voltage divider:

- $V_{DD} = 24$ V. $R_1 = 10\ \text{M}\Omega$ from V_{DD} to the gate, and $R_2 = 6.8\ \text{M}\Omega$ from the gate to ground.
- $R_D = 2\ \text{k}\Omega$ from V_{DD} to the drain, and $R_S = 1\ \text{k}\Omega$ from the source to ground.

a. Find the device constant k. [1 mark]

b. Find V_G. Then find I_{DQ} and V_{GSQ} by solving the quadratic, and justify your choice of root. [3 marks]

c. Find V_{DS} and confirm that the MOSFET is in saturation. [2 marks]

d. Find g_m at the Q-point. If R_S were bypassed and r_d is very large, what would A_v be? [1 mark]

Question 7 [8 marks]

Question 7 circuit: JFET common-source stage RC-coupled to a BJT common-emitter stage

A two-stage RC-coupled amplifier runs from a single supply, $V_{DD} = V_{CC} = 15$ V.

- **Stage 1 (JFET common source):** n-channel JFET with $I_{DSS} = 8$ mA and $V_P = -4$ V (r_d may be ignored). $R_G = 1\ \text{M}\Omega$ from the gate to ground. $R_{S1} = 480\ \Omega$ from the source to ground, fully bypassed. $R_{D1} = 3.3\ \text{k}\Omega$ from the supply to the drain. A DC analysis gives $V_{GSQ} = -1.5$ V (you may use this value).
- **Coupling:** the drain of stage 1 is coupled by a capacitor to the base of stage 2.
- **Stage 2 (BJT common emitter):** NPN BJT with $\beta = 150$ (r_o may be ignored). $R_1 = 33\ \text{k}\Omega$ from the supply to the base, and $R_2 = 10\ \text{k}\Omega$ from the base to ground. $R_{C2} = 2.2\ \text{k}\Omega$ from the supply to the collector. $R_{E2} = 1.06\ \text{k}\Omega$ from the emitter to ground, fully bypassed. A DC analysis gives $I_{E2} = 2.5$ mA (you may use this value).
- The source has $R_{sig} = 10\ \text{k}\Omega$ and drives the gate of stage 1. The collector of stage 2 drives $R_L = 4.7\ \text{k}\Omega$ through a capacitor.

a. Find g_{m1}, r_{e2} and the input impedance of stage 2, Z_{i2}. [2 marks]

b. Find the loaded gain of each stage, A_{v1} and A_{v2}, and the total gain $A_{vT} = v_o/v_i$. [3 marks]

c. Find Z_i and Z_o of the whole amplifier, the overall gain $A_{v_s} = v_o/v_s$, and v_o for $v_s = 2$ mV peak. State the phase of v_o relative to v_s. [2 marks]

d. Stage 1 has a no-load gain of about -8, but its loaded gain is much smaller. Explain why in one sentence, and name one change that would fix it. [1 mark]

Question 8 [7 marks]

a. An op-amp has a differential gain $A_d = 4000$. The input voltages are $V_{i1} = 1.05$ mV and $V_{i2} = 0.95$ mV. Determine the output voltage when the CMRR is (i) 80 dB and (ii) 40 dB. [3 marks]

b. A BJT differential amplifier uses two matched NPN transistors with $\beta = 100$. Each collector connects to $+V_{CC} = +12$ V through $R_C = 10\ \text{k}\Omega$. The two emitters are joined and connected to $-V_{EE} = -12$ V through a single tail resistor $R_E = 15\ \text{k}\Omega$. Both bases are at 0 V DC (the input sources are ground-referenced). The output is taken single-ended from one collector.

Question 8(b) circuit: BJT differential pair with a 15 kOhm tail resistor

Find the DC emitter current of each transistor, r_e and V_C. Then find the single-ended differential gain A_d, the common-mode gain A_c, and the CMRR as a ratio and in dB. [4 marks]

Question 9 [5 marks]

Question 9 circuit: non-inverting amplifier feeding a two-input inverting summer

Two ideal op-amps are powered from ± 15 V.

- **Op-amp 1** is a non-inverting amplifier: $V_1 = 20$ mV rms is applied to its non-inverting input, $R_1 = 4.7\ \text{k}\Omega$ goes from the inverting input to ground, and $R_f = 47\ \text{k}\Omega$ goes from the output to the inverting input.
- **Op-amp 2** is an inverting summing amplifier with its non-inverting input grounded. The output of op-amp 1 connects to op-amp 2's inverting input through $33\ \text{k}\Omega$. A second signal, $V_2 = 50$ mV rms (in phase with V_1), connects to the same inverting input through $10\ \text{k}\Omega$. The feedback resistor of op-amp 2 is $100\ \text{k}\Omega$.

a. Find the output of op-amp 1. [2 marks]

b. Find the output V_o of op-amp 2, with its phase. [2 marks]

c. What input impedance does the source V_2 see? Explain in one line. [1 mark]

Question 10 [7 marks]

a. A series-fed class A amplifier has $V_{CC} = 18$ V, $R_B = 1.2\ \text{k}\Omega$ from V_{CC} to the base, and $R_C = 20\ \Omega$ from V_{CC} to the collector. The emitter is grounded. The transistor has $\beta = 40$. The input signal produces a sinusoidal base current of 8 mA **peak**.

Question 10(a) circuit: series-fed class A amplifier

Find I_{CQ} and V_{CEQ}, the AC output power, the DC input power and the efficiency. [4 marks]

Question 10(b) circuit: class B push-pull, V_CC = 30 V, 16 ohm load

b. A class B push-pull amplifier has $V_{CC} = 30$ V and drives a 16 Ω load. Determine (i) the maximum input power, (ii) the maximum output power and (iii) the maximum circuit efficiency. [3 marks]

END OF PAPER